

Metis FP4 GRPO Training: Experiment Summary and Next Steps

Method: Mean-Bias Quantization Scheme

Core Idea

In standard FP4 quantization, activations x are quantized directly, and the limited dynamic range can hurt precision. The Mean-Bias method reduces dynamic range by subtracting a mean term before quantizing the residual:

$$\hat{x} = \text{Quant}_{\text{FP4}}(x - \mu) + \mu.$$

Here, μ is computed differently in training and inference.

Training Phase

For input shape (B, S, H) , the mean is computed once along the sequence dimensions:

$$\mu = \frac{1}{B \cdot S} \sum_{b,s} x_{b,s,:}.$$

After centering, the residual $x - \mu$ is quantized to FP4, and the quantization scalar is determined by the residual dynamic range.

Inference Phase (Prefill + Decode)

Prefill: with full prompt input of shape $(B, S_{\text{prompt}}, H)$, compute and cache mean once,

$$\mu_{\text{cache}} = \frac{1}{B \cdot S_{\text{prompt}}} \sum_{b,s} x_{b,s,:}.$$

Decode: each step has one new token per batch item, with input shape $(B, 1, H)$. Let $x_{\text{new}} \in \mathbb{R}^{B \times H}$ denote the new token activations (squeezed on sequence dimension), and let n be the number of generated tokens already accumulated in the cache. Update:

$$\mu_{\text{cache}} \leftarrow \frac{n \mu_{\text{cache}} + \sum_{b=1}^B x_{\text{new}}^{(b)}}{n + B}.$$

At each decode step, quantization uses the current μ_{cache} for bias removal before FP4 quantization.

Inference Performance (GSM8K Accuracy)

Method	GSM8K Accuracy
bf16 (baseline)	20%
Mean-Bias FP4	10%
SVD + FP4	2%

Experimental Observations

1) Training Stability Issues

Direct Metis FP4 GRPO training is unstable, with inconsistent outcomes and no reliable convergence.

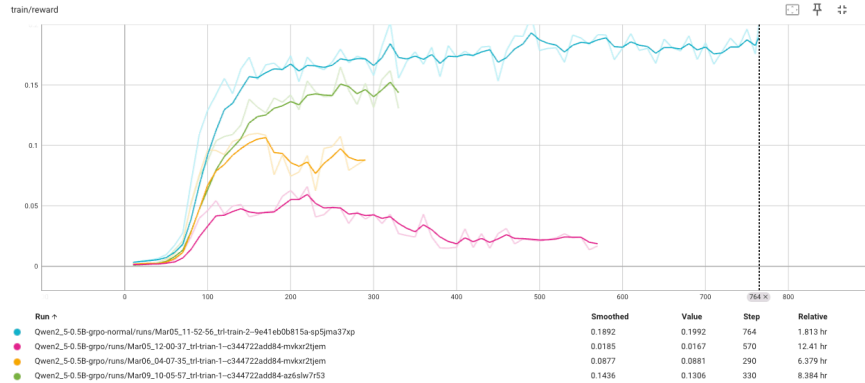


Figure 1: reward

Rollout text quality: beginning as early as step 2, FP4 rollouts show severe degradation patterns: garbled text, meaningless mixed output, repeated symbols/words, and semantically broken yet superficially fluent responses. Since degradation appears at the beginning of training, this indicates insufficient precision in the FP4 path itself.

Reward metrics: FP4 has a much lower reward peak (~ 0.06 vs. ~ 0.2 for bf16). Reward standard deviation first rises and then falls, suggesting weak early learning signals that later collapse.

Clipped ratio: FP4 clipped ratio stays high and rises again, while bf16 quickly approaches zero and stabilizes. This indicates unstable gradient directions at each step and failure to converge to an effective policy, consistent with persistent quantization noise.

Activation statistics: no NaN/Inf is observed, and per-layer `abs_max` remains stable. Numerical overflow is therefore not the root cause.

2) Progress on Quality Evaluation System (A)

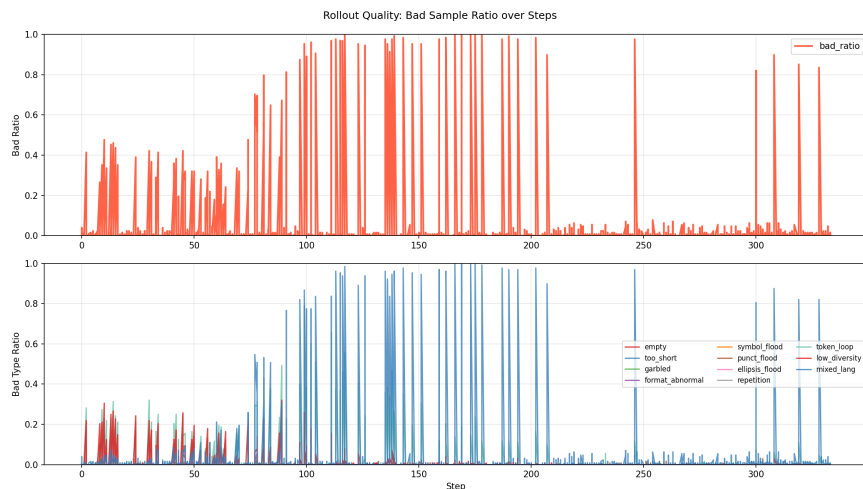


Figure 2: bad_ratio

A rollout quality evaluation framework is in place and can clearly separate bf16 vs. FP4 decode outputs, converting subjective impressions into objective metrics.

3) Rollout-Train Mismatch Findings (B)

Key findings:

- bf16 shows significantly better mismatch behavior than FP4.
- Absolute probability difference is about 0.5 for bf16 vs. about 0.1 for FP4 (with fluctuations).
- This quantitatively captures stronger rollout-train distribution shift under FP4.

Core Hypothesis

Insufficient FP4 forward precision (degradation visible from step 2) causes severe early rollout degradation. GRPO then reinforces “less bad” responses within groups with very small reward variance. Quantization noise continuously perturbs gradient direction (high, oscillating clipped ratio), preventing convergence and eventually weakening the model’s ability to distinguish quality.

Two Key Problems

- **Rollout quality:** using FP4 decode directly during bf16 post-training causes quality drop.

- **Rollout-train mismatch:** current mean-bias rollout uses prefix-based mean (cannot access future tokens), while training computes mean over all tokens jointly.

Next-Step Plan

C. Step-Wise Position and Mean Analysis

Core questions:

1. How representative is prefill mean for full-sentence mean?
 - Analyze trajectories of prefill mean, decode-time mean, and training forward mean.
 - Evaluate cumulative mean-estimation error as generation length grows.
2. Where does degradation begin?
 - Track token-wise perplexity and top- k distribution gap.
 - Determine whether divergence starts from early tokens or collapses after mid/late accumulated error.

D. FP4 Pre-Adaptation (QAT + Self-Distillation)

Core idea: before FP4 RL training, perform QAT-style self-distillation. Otherwise, RL must both explore and recover FP4 quality, which is too heavy.

- **Goal:** adapt model to FP4 quantization before RL.
- **Method:** distill from bf16 teacher to FP4 student using logits KL + CE warmup.
- **Evaluation:** compare pure decode quality before/after warmup.
- **Expectation:** narrow bf16-FP4 gap and provide a better RL starting point.

E. RL Anti-Noise Mechanisms (Later Stage)

- Filter or down-weight abnormal rollouts.
- Tune clip range, learning rate, and KL penalty.
- Monitor whether clipped ratio returns to a stable regime.

Summary

The central issue in FP4 GRPO training is insufficient quantization precision, which leads to unstable training and poor convergence reliability. The quality evaluation system (A) now clearly distinguishes bf16 and FP4 behavior, and mismatch analysis (B) quantifies the distribution shift (approximately 0.5 vs. 0.1).

Next, we will focus on **C** (mean representativeness and degradation localization) and **D** (QAT self-distillation pre-adaptation). The guiding strategy is to make the model adapt to FP4 first, then enter RL training with a better optimization starting point.